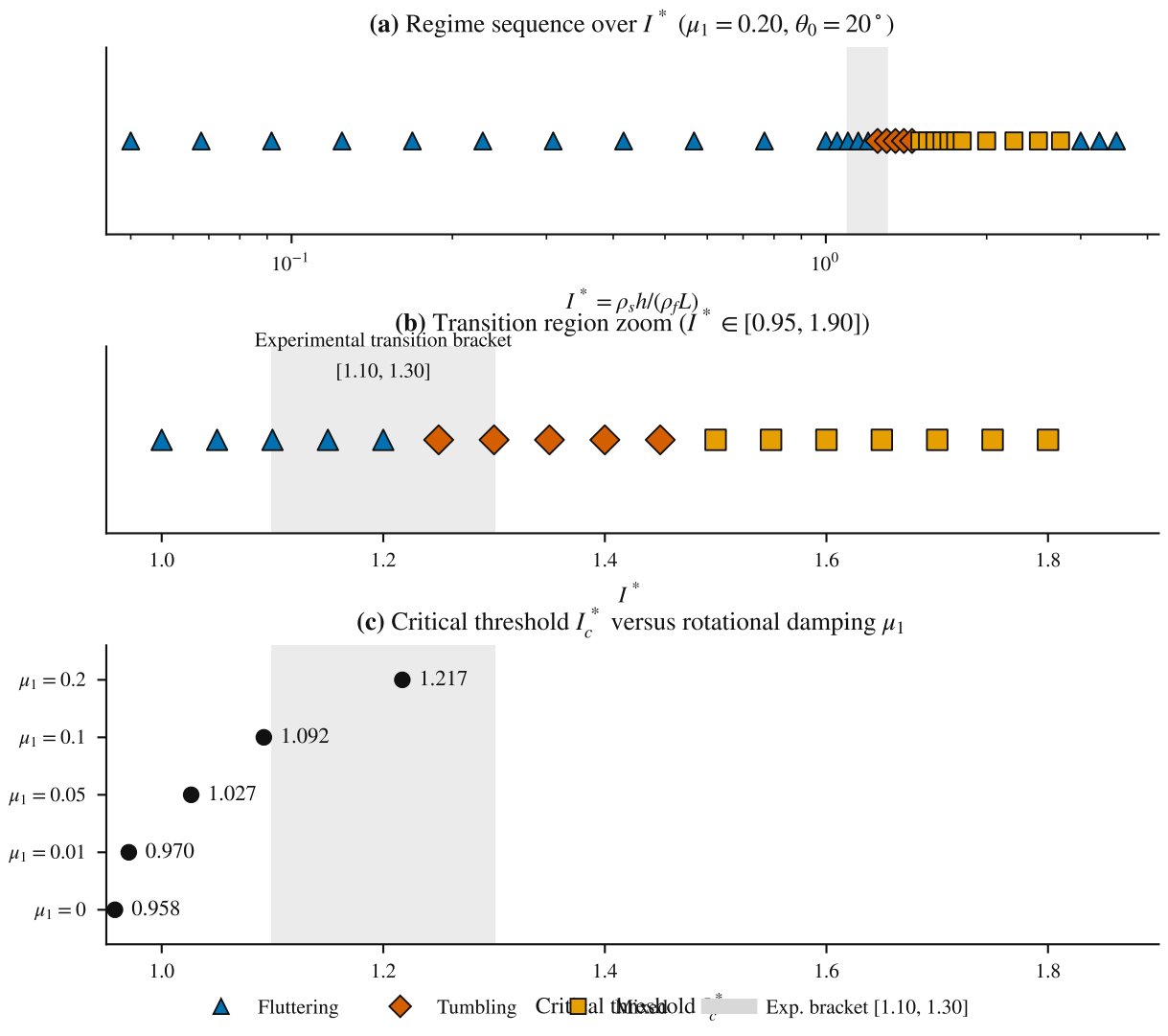
(a) Regime sequence over I^* ($\mu_1 = 0.20$, $\theta_0 = 20^\circ$)
10^{-1}
10^{0}
$I^* = \rho_s h/(\rho_f L)$
(b) Transition region zoom ($I^* \in [0.95, 1.90]$)
Experimental transition bracket
[1.10, 1.30]
1.0
1.2
1.4
1.6
1.8
I^*
(c) Critical threshold I_c^* versus rotational damping μ_1
$\mu_1 = 0.2$
1.217
$\mu_1 = 0.1$
1.092
$\mu_1 = 0.05$
1.027
$\mu_1 = 0.01$
0.970
$\mu_1 = 0$
0.958
1.0
1.2
1.4
1.6
1.8
Fluttering
Tumbling
Critical threshold I_c^*
Exp. bracket [1.10, 1.30]